

ChatKid: A Child-Centered AI Chatbot Prototype for Supporting Children’s Questions with Parent-Guided Support

MEHERUNNESA TANIA*, Boise State University, USA

JERRY ALAN FAILS*, Boise State University, USA

Children increasingly ask AI chatbots, learning applications, and digital toys a wide range of questions, from why the sky is blue to questions about emotions, conflict, or safety. Yet many systems either answer directly or refuse to respond, without creating space for parent or caregiver involvement. We present *ChatKid*, a child-centered AI chatbot prototype that supports children’s everyday questions and redirects sensitive ones toward parent-guided support. *ChatKid* was shaped by participatory design sessions with children ages 6–11. Each question first passes a rule-based safety filter: allowed questions receive a child-friendly answer, while sensitive questions are redirected to a parent-guided flow instead of being answered by the model. Through this design, *ChatKid* positions AI as a supportive helper rather than a replacement for human guidance. This work contributes an early-stage prototype and design direction for child–parent–AI interaction that supports children’s curiosity, safety, and family guidance.

CCS Concepts: • **Human-centered computing** → **Empirical studies in interaction design**.

Additional Key Words and Phrases: Children, Artificial Intelligence, Safety, Boundary, Parent Involvement

1 Introduction and Background

Artificial Intelligence (AI) is becoming part of many technologies children use daily, including chatbots, educational applications, digital toys, and interactive media [12–14]. These technologies can support curiosity, learning, and play, but children may not understand how AI works, when it may be wrong, or when to ask a parent for help [3, 10]. Because children can form social and emotional connections with interactive systems [1, 5, 8], AI may be seen not only as a tool, but also as a helper or social partner, raising design challenges around overtrust, safety, boundaries, and family involvement.

Prior child–AI systems show that AI can support children when designed as part of parent–child activities. *StoryBuddy* [15] supports parent–child storytelling, and *BrickSmart* [9] supports family block play and spatial language learning, connecting to Joint Media Engagement, which emphasizes that children benefit from using media with others [7]. When children use AI alone, however, they may overtrust confident responses [11] and need help recognizing AI’s limitations [2], making adult guidance especially important around sensitive or uncertain topics.

Existing systems often address one part of this problem, such as AI-supported learning, AI literacy, or parent involvement, with less attention to combining these in a child-centered chatbot that makes AI boundaries visible and routes sensitive topics toward parent support. We present *ChatKid*, a chatbot prototype shaped by participatory design sessions with children. The prototype uses a rule-based safety filter to answer allowed questions in child-friendly language and redirect sensitive questions toward parent-guided support. This work contributes a preliminary design case showing how children’s own judgments about which questions belong with AI versus parents can inform boundaries in family-centered AI systems.

*Both authors contributed equally to this research.

Authors’ Contact Information: [Meherunnesa Tania](mailto:meherunnesatania@boisestate.edu), meherunnesatania@boisestate.edu, Boise State University, Boise, Idaho, USA; [Jerry Alan Fails](mailto:jerryfails@boisestate.edu), jerryfails@boisestate.edu, Boise State University, Boise, Idaho, USA.

2 Co-Designing and Developing *ChatKid*

Building on this design opportunity, this work asks: *How can a child-centered AI chatbot support children’s everyday questions while making AI boundaries visible and encouraging parent involvement for sensitive topics?* *ChatKid* explores this question by combining AI support, safety boundaries, and parent-guided mediation within one interaction flow.

ChatKid is guided by three design goals: (1) support children’s question-asking, (2) distinguish AI-appropriate topics from topics that may need parent involvement, and (3) encourage parent–child interaction around sensitive questions. Rather than simply blocking sensitive topics, *ChatKid* shows a boundary message and prompts the child to involve a parent when needed. The prototype was shaped by participatory design activities with children, described next.

2.1 Participatory Design

ChatKid was shaped through four participatory design sessions with seven children ages 6–11, following a child-centered approach inspired by cooperative inquiry [4, 6]. The sessions were reviewed by our university’s IRB (IRB #126-SB23-026), and parents provided written consent for participation and use of session photographs. We explored how children think about AI, what they expect from a chatbot, and when they believe a chatbot should involve a parent. Parents were not part of these sessions, a limitation we return to in Section 4.

Children completed a Question Jar activity, sorting 30 question cards by who they would prefer to ask: a *Parent*, an *AI Agent*, *Not Sure*, or *Neither*. Facilitators grouped children’s reasons inductively; children favored AI for factual or curiosity-based questions and parents for emotional, personal, or sensitive ones.

In the next two sessions, children co-designed three chatbot concepts in small groups and then merged them into one shared *ChatKid* concept. Their ideas included voice input, customizable avatars, mood-based visuals, and parent-support requests. Finally, children tested a working prototype (Section 3) and gave one-idea-per-note feedback. They liked the playful interface, avatars, backgrounds, and typing or voice support, and asked for more avatar choices, emoji and reading support, additional voices, and better handling of flagged messages. Together, these sessions shaped *ChatKid*’s interface, safety boundaries, and parent-support flow.

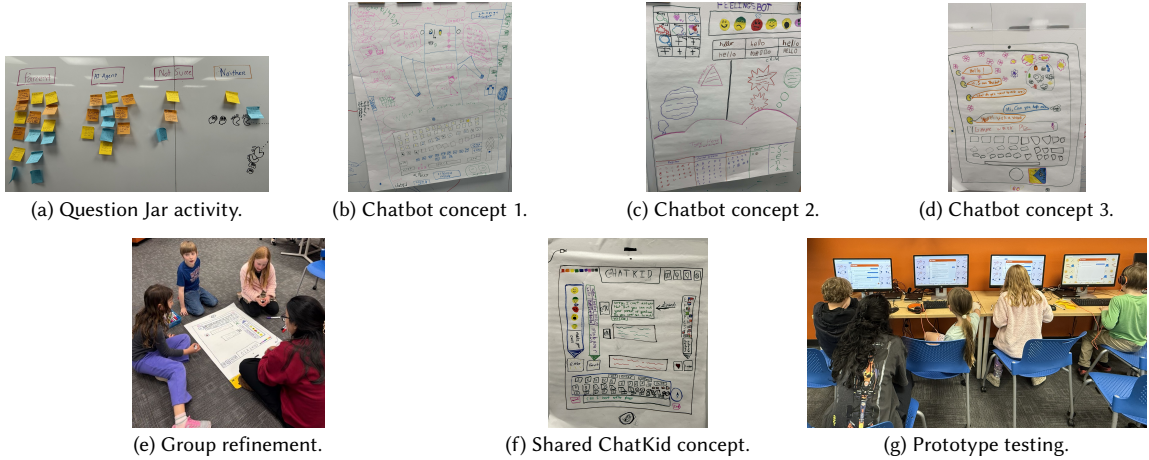


Fig. 1. Children’s participatory design activities that shaped the design of *ChatKid*.

3 System Architecture and Implementation

ChatKid is a locally hosted web-based chatbot prototype built with HTML, CSS, and JavaScript. It connects to a local Ollama API and uses the llama3.2:1b model to generate responses, keeping all interaction data on-device rather than sent to a third-party service. As shown in Figure 2, each question first passes through a rule-based safety filter. Allowed questions receive a child-friendly response from the local language model; restricted questions are not sent to the model and are instead redirected to a parent-guided support flow, keeping sensitive topics within a safer interaction path.

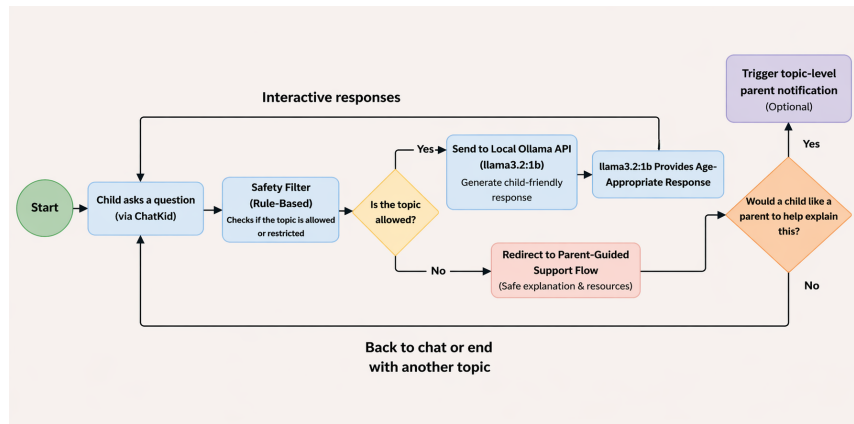


Fig. 2. System workflow of *ChatKid*.

3.1 Safety Filtering and Response Generation

Before a child’s message reaches the language model, *ChatKid* checks it against a list of restricted keywords covering harm, violence, bullying, emotions, unsafe situations, privacy, abuse, and inappropriate content. A keyword list was used because it is simple to inspect and adjust while the prototype is still early.

Messages that pass the check go to the locally hosted llama3.2:1b model, small enough to run entirely on one machine. A system prompt asks the model to answer as *ChatKid* in simple, kind, and short language, letting children ask anything rather than choose from a script – a child can ask why the sky is blue and get an age-appropriate answer. Figure 3 shows the interface and an example.

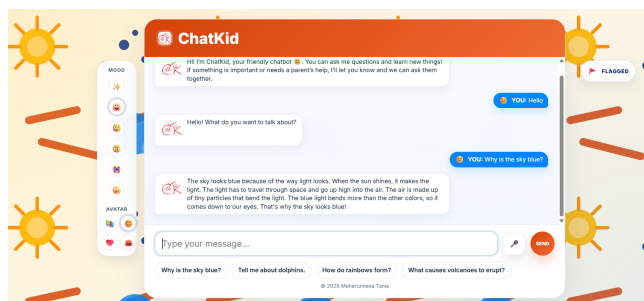
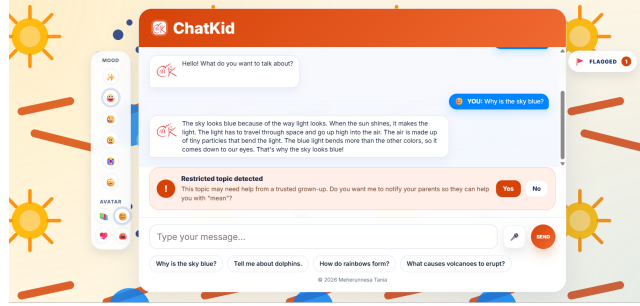


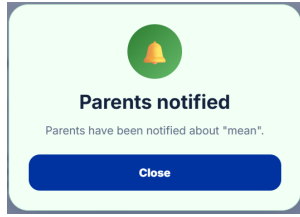
Fig. 3. *ChatKid* interface showing the chat area, text input box, microphone button, suggestion prompts, mood options, avatar choices, and flagged-message feature.

3.2 Boundary Explanation and Parent Mediation

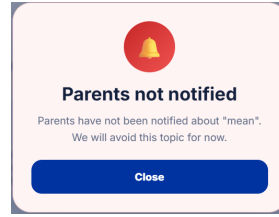
When a message is flagged, *ChatKid* does not answer it. It shows a short message about why, then asks if the child would like help from a parent or guardian. If the child says yes, the topic is marked as parent support requested; if no, *ChatKid* notes that and drops the topic. Either way, the question is saved in the flagged-message view along with the triggering keyword and whether a parent was asked, keeping the handling of sensitive topics visible.



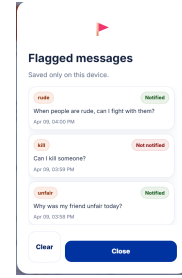
(a) Restricted-topic prompt.



(b) Parent support requested.



(c) Parent support not requested.



(d) Flagged messages.

Fig. 4. Parent-support request outcomes and flagged-message view in *ChatKid*.

4 Limitations and Future Work

ChatKid is an early prototype shaped mainly by children’s perspectives; parents did not participate, so the parent-guided flow reflects children’s expectations rather than parents’ input. The keyword filter is simple and inspectable, but may miss sensitive questions or flag harmless ones. The small local model may also produce brief or incomplete answers. Future work should explore context-aware filtering, stronger response generation, and privacy-conscious parent-support features with families before longer-term home testing. Because the sessions involved only seven children, the findings are preliminary design insights rather than generalizable conclusions.

5 Conclusion

ChatKid supports children’s everyday questions while redirecting sensitive ones toward parent or caregiver support. Children ages 6–11 shaped the prototype through participatory design, and their sorting of questions — AI for facts, parents for feelings — informed the system boundary. We offer *ChatKid* as an early direction for child-centered AI: a helper that knows when to step aside. This work raises open questions about how much visibility parents should have into flagged questions and who should decide that boundary: the child, parent, or system.

References

- [1] Bradley J. Bond and Sandra L. Calvert. 2014. A model and measure of US parents' perceptions of young children's parasocial relationships. *Journal of Children and Media* 8, 3 (2014), 286–304. doi:10.1080/17482798.2014.890948
- [2] Aayushi Dangol, Runhua Zhao, Robert Wolfe, Trushaa Ramanan, Julie A. Kientz, and Jason Yip. 2025. "AI just keeps guessing": Using ARC Puzzles to Help Children Identify Reasoning Errors in Generative AI. In *Proceedings of the 24th Interaction Design and Children (IDC '25)*. Association for Computing Machinery, New York, NY, USA, 444–464. doi:10.1145/3713043.3728836
- [3] Stefania Druga, Randi Williams, Cynthia Breazeal, and Mitchel Resnick. 2017. "Hey Google is it OK if I eat you?": Initial Explorations in Child-Agent Interaction. In *Proceedings of the 2017 Conference on Interaction Design and Children* (Stanford, California, USA) (IDC '17). Association for Computing Machinery, New York, NY, USA, 595–600. doi:10.1145/3078072.3084330
- [4] Allison Druin. 1999. Cooperative inquiry: developing new technologies for children with children. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Pittsburgh, Pennsylvania, USA) (CHI '99). Association for Computing Machinery, New York, NY, USA, 592–599. doi:10.1145/302979.303166
- [5] James H. Gray, Emily Reardon, and Jennifer A. Kotler. 2017. Designing for Parasocial Relationships and Learning: Linear Video, Interactive Media, and Artificial Intelligence. In *Proceedings of the 2017 Conference on Interaction Design and Children* (Stanford, California, USA) (IDC '17). Association for Computing Machinery, New York, NY, USA, 227–237. doi:10.1145/3078072.3079736
- [6] Mona Leigh Guha, Allison Druin, Gene Chipman, Jerry Alan Fails, Sante Simms, and Allison Farber. 2004. Mixing ideas: a new technique for working with young children as design partners. In *Proceedings of the 2004 Conference on Interaction Design and Children: Building a Community* (Maryland) (IDC '04). Association for Computing Machinery, New York, NY, USA, 35–42. doi:10.1145/1017833.1017838
- [7] Alexis Hiniker, Amelia Wang, Jonathan Tran, Mingrui Ray Zhang, Jenny Radesky, Kiley Sobel, and Sungsoo Ray Hong. 2021. Can Conversational Agents Change the Way Children Talk to People?. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference* (Athens, Greece) (IDC '21). Association for Computing Machinery, New York, NY, USA, 338–349. doi:10.1145/3459990.3460695
- [8] Donald Horton and R. Richard Wohl. 1956. Mass communication and parasocial interaction: Observations on intimacy at a distance. *Psychiatry* 19, 3 (1956), 215–229. doi:10.1080/00332747.1956.11023049
- [9] Yujia Liu, Siyu Zha, Yuewen Zhang, Yanjin Wang, Yangming Zhang, Qi Xin, Lun Yiu Nie, Chao Zhang, and Yingqing Xu. 2025. BrickSmart: Leveraging Generative AI to Support Children's Spatial Language Learning in Family Block Play. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 645, 19 pages. doi:10.1145/3706598.3714212
- [10] Duri Long and Brian Magerko. 2020. What is AI Literacy? Competencies and Design Considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3313831.3376727
- [11] Luis Morales-Navarro, Phillip Gao, Eric Yang, and Yasmin B Kafai. 2024. "It's smart and it's stupid." Youth's conflicting perspectives on LLMs' language comprehension and ethics. In *Proceedings of the 19th WiPSCe Conference on Primary and Secondary Computing Education Research* (Munich, Germany) (WiPSCe '24). Association for Computing Machinery, New York, NY, USA, Article 36, 2 pages. doi:10.1145/3677619.3678131
- [12] UNICEF. 2021. *Policy Guidance on AI for Children*. Technical Report. UNICEF Office of Global Insight and Policy. <https://www.unicef.org/innocenti/media/1341/file/UNICEF-Global-Insight-policy-guidance-AI-children-2.0-2021.pdf> Version 2.0.
- [13] UNICEF. 2025. *Guidance on AI and Children*. Technical Report. UNICEF Innocenti – Global Office of Research and Foresight. <https://www.unicef.org/innocenti/media/11991/file/UNICEF-Innocenti-Guidance-on-AI-and-Children-3-2025.pdf> Version 3.0.
- [14] Ying Xu and Mark Warschauer. 2020. What Are You Talking To?: Understanding Children's Perceptions of Conversational Agents. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376416
- [15] Zheng Zhang, Ying Xu, Yanhao Wang, Bingsheng Yao, Daniel Ritchie, Tongshuang Wu, Mo Yu, Dakuo Wang, and Toby Jia-Jun Li. 2022. StoryBuddy: A Human-AI Collaborative Chatbot for Parent-Child Interactive Storytelling with Flexible Parental Involvement. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 218, 21 pages. doi:10.1145/3491102.3517479